JR Validated Environment — Curve Module

User Guide for XY Curve Analysis

Version 1.0
2026-03-26
JR Validated Environment

1. Introduction

The Curve Module provides one validated Python script, jrc_curve_properties, for extracting engineering-relevant properties from an XY measurement series. Typical applications include force-displacement curves, torque-angle profiles, pressure-volume cycles, and any other physical measurement captured as a sequence of (X, Y) pairs.

Unlike a mathematical function, the input is treated as a time-ordered measurement series. The same X value may appear multiple times — for example, both the loading and unloading arms of a compression test pass through the same displacements. Named phases resolve this ambiguity by anchoring curve segments to specific row ranges within the data.

The script is driven entirely by a plain-text configuration file (INI format). You select which properties to compute by adding the appropriate keys to the config file. Absent keys are silently skipped. Results are printed to the terminal and written to a text file. An optional PDF plot annotates the computed properties on the raw curve.

The script runs through jrrun, which verifies project integrity and ensures the correct Python environment is used on every execution.

> i **Validated environment:** Never run jrc_curve_properties.py directly with python. Always use the wrapper or jrrun so that the integrity check and controlled library are applied.

2. Usage

The script accepts a single argument: the path to a .cfg configuration file.

```
jrc_curve_properties path/to/config.cfg
jrc_curve_properties --help
```

All file paths within the config (data CSV, output files) are resolved relative to the directory containing the config file, not the working directory. This means you can invoke the script from any directory as long as the config path is correct.

> i **Tip:** Keep the config file in the same folder as your data CSV. The results file and plot PDF will then appear alongside them by default.

3. Data File Format

The input is a two-column CSV file with a header row. The column names can be anything; you specify them in the [data] section of the config file.

Example

```
displacement_mm,force_N
0.0,0.0
0.5,12.4
1.0,25.1
...
```

- Column names are case-sensitive.
- Decimal separator must be '.' regardless of system locale.

- Rows are processed in file order (time order). Do not sort by X.
- Additional columns are ignored.

⚠ **Row order matters:** The script treats the data as a time-ordered sequence. If your measurement file is sorted by X instead of by measurement time, phase extraction will not behave as expected. Keep the original acquisition order.

4. Config File Reference

The config file uses standard INI format. Section names are case-sensitive. Keys use dot notation for modifiers (e.g. secant.x1, y_at_x_1.phase). Numbered suffixes (_1, _2, …) allow multiple instances of the same query or transition type in a single run.

ℹ **Config rules:** Absent key = feature is skipped entirely. All numeric values use '.' as the decimal separator. Boolean values are 'yes' or 'no'.

4.1 [data] (required)

Specifies the data source. This section is mandatory.

Key	Values	Description
file	path	Path to the CSV file, relative to the config file directory.
x_col	string	Name of the column to use as X (e.g. displacement_mm).
y_col	string	Name of the column to use as Y (e.g. force_N).

4.2 [output] (optional)

Controls labels, plot generation, and output file paths.

Key	Values	Description
title	string	Title shown on the plot and in the results file header.
label_x	string	X-axis label on the plot.
label_y	string	Y-axis label on the plot.
plot	yes | no	Generate a PDF plot. Default: no.
plot_file	path	Output path for the PDF plot. Default: <cfg>_plot.pdf.
results_file	path	Output path for the results text file. Default: <cfg>_results.txt.

4.3 [smoothing] (optional)

Sets the default smoothing method used for all derivative-based calculations (instantaneous slope, inflection points, yield point). Smoothing is NEVER applied to raw statistics: max/min, AUC, and hysteresis always use raw data. Individual phases can override these settings with smooth_method and smooth_span in their [phase.NAME] section.

Key	Values	Description
method	savgol | moving_avg | none	Smoothing algorithm. savgol (Savitzky-Golay) is recommended for preserving peak positions. Default: savgol.
span	0.0 – 1.0	Fraction of the data length used as the smoothing

		window. Larger = more smoothing. Default: 0.15.
apply_to_plot	yes | no	Overlay the smoothed curve on the plot for each phase. Default: no.

✅ **Choosing span:** A good starting value is 0.15 for general smoothing and 0.30 for inflection detection. Too little smoothing leaves d2y noisy; too much shifts the reported inflection. Use the [debug] d2y output to inspect the second derivative and find a stable value. See Section 7 for the tuning workflow.

4.4 [phase.NAME] (optional, repeatable)

Defines a named segment of the data by anchoring start and end boundaries to physical X values. The phase name becomes part of the output label (e.g. max Y [loading]). Define as many phases as needed; they are processed in the order they appear in the config file.

Key	Values	Description
x_start	number	X value marking the start of this phase. Matched to the nearest data point in the search zone.
x_end	number	X value marking the end of this phase. Matched to the nearest data point at or after x_start.
search	ascending | descending	Restrict the search to one arm of a peak/valley curve. 'ascending' searches only the ascending arm (up to the X maximum in the zone); 'descending' searches only the descending arm. Omit to search the full zone.
after_phase	phase name	Restrict the search zone to rows after the named phase ends. The referenced phase must be declared earlier in the config file.
max_y	yes	Report the maximum Y within this phase and its X position.
min_y	yes	Report the minimum Y within this phase and its X position.
max_x	yes	Report the maximum X within this phase and its Y position.
min_x	yes	Report the minimum X within this phase and its Y position.
auc	yes	Area under this phase (trapezoidal rule, raw data).
smooth_method	savgol | moving_avg | none	Override the global smoothing method for derivative calculations in this phase.
smooth_span	0.0 – 1.0	Override the global smoothing span for this phase.

Example: a loading/unloading curve where both arms pass through the same X values.

```
[phase.loading]
x_start = 0
x_end   = 80
search  = ascending
max_y   = yes
auc     = yes
```

```
[phase.unloading]
x_start       = 80
x_end         = 0
search        = descending
after_phase   = loading
min_y         = yes
```

4.5 [global] (optional)

Properties computed over the full dataset (or over a named phase via the auc.phase modifier). For phase-scoped max/min/AUC it is more natural to use keys directly in the [phase.NAME] section.

Key	Values	Description
max_y	yes	Maximum Y over the full dataset.
min_y	yes	Minimum Y over the full dataset.
max_x	yes	Maximum X over the full dataset.
min_x	yes	Minimum X over the full dataset.
auc	yes	Area under the full curve (trapezoid rule, raw data).
auc.phase	phase name	Restrict AUC to a named phase.
hysteresis	yes	Area between two arms (linear interpolation, raw data). Requires loading and unloading phases.
hysteresis_loading.phase	phase name	Phase to use as the loading arm (default: loading).
hysteresis_unloading.phase	phase name	Phase to use as the unloading arm (default: unloading).

4.6 [slope] (optional)

Three types of slope are available. All can be scoped to a named phase via a .phase modifier. Multiple at_x queries are supported with numbered suffixes (at_x_1, at_x_2, …).

Key	Values	Description
overall	yes	Linear regression slope and R^2 over the dataset or phase.
overall.phase	name	Phase name for the overall slope.
overall.plot	yes	Draw the regression line on the plot.
secant	yes	Slope of the secant line between secant.x1 and secant.x2.
secant.x1	number	First X value for the secant.
secant.x2	number	Second X value for the secant.
secant.phase	name	Phase name for the secant.
secant.plot	yes	Draw the secant line on the plot.
at_x_1	number	X value for instantaneous slope (numerical derivative, smoothed data). Add at_x_2, at_x_3, … for additional points.
at_x_1.phase	name	Phase name for this at_x query.
at_x_1.plot	yes	Draw the tangent line at this point on the plot.

4.7 [query] (optional)

Interpolated lookups. All three query types support numbered suffixes (_1, _2, …) to run multiple queries in one pass.

Key	Values	Description
y_at_x_1	number	X value — returns interpolated Y at that X.
y_at_x_1.phase	name	Phase name (omit for full dataset).
y_at_x_1.show	yes	Draw a + marker on the plot at this point.
x_at_y_1	number	Y value — returns X where the curve crosses that Y.
x_at_y_1.phase	name	Phase name (omit for full dataset).
x_at_y_1.mode	first | last | all	Which crossing(s) to return (default: first).
y_at_rel_x_1	number (x_ref)	Returns Y at the point x_ref × (1 + frac).
y_at_rel_x_1.frac	number	Relative offset fraction (e.g. 0.1 = +10% of x_ref, −0.1 = −10% of x_ref). Default: 0.
y_at_rel_x_1.phase	name	Phase name (omit for full dataset).
y_at_rel_x_1.show	yes	Draw a + marker on the plot at this point.

4.8 [transform] (optional)

Applied to Y values after loading and before all computations and plotting. When both keys are present, y_scale is applied first, then y_offset_x. Both transforms are recorded in the results file.

Key	Values	Description
y_scale	number	Multiply every Y value by this factor. Use for unit conversion or normalisation.
y_offset_x	number (X value)	Subtract the Y value at this X from every Y in the dataset. Use to zero the curve at a reference point (e.g. the start of a phase).

> **Label update:** After applying y_scale, update label_y in [output] to reflect the new units.

4.9 [transitions] (optional)

Detects structural changes in the curve: inflection points (sign change of the second derivative of the smoothed curve) and yield-like points (where the slope drops to a specified fraction of its maximum). Both use the smoothed data from [smoothing] or the per-phase override.

Inflection points

Key	Values	Description
inflections	yes	Detect inflections in the dataset or named phase.
inflections.phase	name	Phase name (omit for full dataset).
inflections.min_gap	number	Minimum X spacing between reported inflections. Default: 5% of the phase X range. Set to 0 to report all sign changes.
inflections.plot_slope	yes	Draw a tangent line at each inflection on the plot.

To run inflection detection on multiple phases, use numbered suffixes (_1, _2, …):

```
inflections_1             = yes
inflections_1.phase       = loading
inflections_1.plot_slope  = yes
inflections_2             = yes
inflections_2.phase       = unloading
```

Yield point

Key	Values	Description
yield.slope	0.0 – 1.0	Fraction of max slope at which to report the yield-like point. E.g. 0.10 = where slope drops to 10% of its maximum.
yield.phase	name	Phase name (omit for full dataset).
yield.show	yes	Draw an × marker on the plot at the yield point.

To detect yield points in multiple phases, use numbered suffixes (_1, _2, …):

```
yield_1.slope  = 0.10
yield_1.phase  = loading
yield_1.show   = yes
yield_2.slope  = 0.10
yield_2.phase  = unloading
yield_2.show   = yes
```

4.10 [debug] (optional)

Writes diagnostic CSV files to help tune smoothing for inflection detection.

Key	Values	Description
d2y	yes	Write a CSV with the smoothed curve and second derivative for the specified phase.
d2y.phase	name	Phase to debug (required when d2y = yes).

Output file: <cfg>_debug_d2y_<phase>.csv. Columns: x, y_raw, y_smooth, d2y, trimmed. The trimmed column is 1 for rows excluded from the inflection search by the boundary trim (half the smoothing window from each end of the phase). Plot x vs d2y filtered to trimmed=0 to see exactly what the inflection algorithm sees.

5. Output Files

File	When created	Contents
<cfg>_results.txt	Every run	All computed properties, aligned in columns. Records raw vs. smoothed source for each value.
<cfg>_plot.pdf	plot = yes in [output]	Plot of raw data with phase boundaries, smoothed curve (if apply_to_plot = yes), and annotated properties. Markers shown only when .show = yes.
<cfg>_debug_d2y_<phase>.csv	d2y = yes in [debug]	Diagnostic CSV: x, y_raw, y_smooth, d2y, trimmed. Use to tune smooth_span for inflection detection.

> **Default names:** Output file names default to the config file stem plus a suffix. Override with results_file and plot_file in [output].

6. Worked Example: Force-Displacement Curve

The following example analyses a loading/unloading force-displacement test. The data file has two columns: displacement_mm and force_N. The test applies force to 80 mm, then releases. Both arms pass through the same displacement values.

Config file: analysis.cfg

```
[data]
file    = force_displacement.csv
x_col   = displacement_mm
y_col   = force_N

[output]
title   = Force-Displacement Analysis
label_x = Displacement (mm)
label_y = Force (N)
plot    = yes

[phase.loading]
x_start = 0
x_end   = 80
search  = ascending
max_y   = yes
auc     = yes

[phase.unloading]
x_start     = 80
x_end       = 0
search      = descending
after_phase = loading
min_y       = yes

[global]
hysteresis = yes

[slope]
overall        = yes
overall.phase  = loading
at_x_1         = 10
at_x_1.phase   = loading
at_x_1.plot    = yes
```

```
[query]
y_at_x_1       = 40
y_at_x_1.phase = loading
y_at_x_1.show  = yes

[transitions]
yield_1.slope = 0.10
yield_1.phase = loading
yield_1.show  = yes
```

Run

```
jrc_curve_properties analysis/analysis.cfg
```

What to look for in the output

- max Y [loading] — peak force during loading, with its displacement.
- AUC [loading] — energy absorbed during loading (N·mm).
- Hysteresis — energy dissipated = loading AUC minus unloading AUC.
- slope overall [loading] — mean stiffness (N/mm) with R^2.
- slope at x=10 [loading] — instantaneous stiffness at 10 mm.
- Y at x=40 [loading] — interpolated force at mid-stroke.
- yield point [loading] — displacement where stiffness drops to 10% of maximum.

7. Tuning Smoothing for Inflection Detection

Inflection points are detected by finding sign changes in the second derivative (d2y) of the smoothed curve. The smooth_span parameter controls how much smoothing is applied before d2y is computed. Getting this right is a two-step process.

Step 1 — Enable debug output

```
[smoothing]
method = savgol
span   = 0.30

[phase.loading]
x_start = 0
x_end   = 80
search  = ascending

[debug]
d2y       = yes
d2y.phase = loading
```

Step 2 — Inspect the d2y CSV

Open the <cfg>_debug_d2y_loading.csv and plot x vs d2y (filter to trimmed = 0). Count the zero-crossings. Each crossing will become a reported inflection.

- Too many crossings → increase span (more smoothing).

- Reported X does not match expected location → decrease span (less smoothing).
- Stability check: if the reported inflection X moves by less than your acceptable tolerance when span changes by ±0.10, the result is reliable.

✅ **Starting values:** span = 0.15 for general slope calculations. span = 0.30 for inflection and yield detection. For very noisy data, span = 0.40 or higher may be needed.

8. Error Messages

Message fragment	Likely cause and action
Unknown section	A section name in the config is not recognised. Check spelling. Valid sections: [data], [output], [smoothing], [global], [slope], [query], [transform], [transitions], [debug], [phase.NAME].
Unknown key	A key inside a known section is not recognised. Check dot-notation modifiers and spelling. Keys are case-sensitive.
must be numeric	A numeric key (e.g. y_scale, secant.x1) has a non-numeric value.
d2y.phase is required	d2y = yes was set in [debug] but d2y.phase was not specified. Add d2y.phase = <phase_name>.
x_col / y_col	A required key is missing from [data]. All three keys (file, x_col, y_col) must be present.
No such file or directory	The data CSV or config file path is incorrect. Verify relative paths are relative to the config file's directory.

9. Validated Environment Notes

jrc_curve_properties is validated under OQ document JR-VR-CURVE-001. It requires the JR environment to be installed and the project integrity file to be current. Any modification to the script or its dependencies requires a change assessment before re-use in a validated context.

The script uses scipy for Savitzky-Golay smoothing (scipy.signal.savgol_filter). The package version is pinned in admin/python_requirements.txt and installed from the local repository at each environment rebuild.

⚠️ **Change control:** Do not modify jrc_curve_properties.py without first raising a change request and re-running admin_curve_oq to confirm that all 28 OQ test cases still pass.